Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

DRAFT — UE26EC141A · Unit_01 · Unit-1 Semiconductor Diode · Parallel and Series- Parallel configurations

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Parallel and Series- Parallel configurations

1 Parallel and Series- Parallel configurations

Study focus: Parallel and Series- Parallel configurations. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000054 · EV0000055

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today, we'll see parallel diode configuration.

Key details

- In the last class, we have solved the numerical and serial diode configuration.
- Today, completely, we look at the concept of parallel diode configuration with the numerical examples.
- The first question here is: Determine the Vout,.
- ID, ID1, and ID2 for the parallel diode configuration of the figure.
- I1 is this, ID2 is this, ID2 is this for the parallel diode.
- Since we have two diodes here connected in parallel, we call it the parallel diode configuration.
- Coming to the solution now.
- Then, how to calculate I1?
- The voltage applied between these two points, it is E minus.
- The applied voltage here and this voltage is 0.7.
- We get 10 minus 0.7, which is 9.3 divided by 0.33.
- Kelvin, which is 28.18.
- Coming to the ID1 and ID2, the potentials are same, and hence ID1 is equal to.
- Current also will be same, so I1 by I2, I1 by 2.
- So both current will be same now, because of the potential is same.
- 28.18 divided by 2, we get 14.09 milliamps.
- In case if the numerical has one series and one parallel, the previous one was both parallel.
- How to go about?
- Then we apply the KVL here.

- The diode is represented by 0.7 VD1, VD2.
- We are also able to apply the KVL this loop also.
- So first go with this.
- Then E applied voltage, leaving end is positive here.
- Then VD1, minus VD1 here, leaving end is minus V2, VD1, and leaving end is plus here, plus minus or the minus here.
- Then calculate the VD2 here.
- After substituting, we get 18.6 volts.
- Then calculate I2 across this.
- Since we know the voltage V2, calculate the value of the I2 with the resistance R2, we get 3.32.
- Then how to calculate this I1?
- The applied voltage here is 0.7.
- The 0.7 by 3.2 kiloohm will give you 0.212 milliamps.
- One thing we need to notice, each numerical may not be same.
- Ohm's law, then we'll be able to get the answer.
- Coming to one more numerical, where diodes are connected in process.
- We have a red LED connected 1.8.
- I think this numerical we solved.
- Just to refresh to people, the diode can be connected in series.
- We already solved this in topic four or five.

Example or application from the lecture

- Given, if you measure the voltage from this point to this point, this is 0.7.
- Coming to the question number three or the second numerical example for series parallel diode.
- Then the total current given by ID2 plus I1 is equal to I2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 05:00–09:59

When applied voltage is positive here, so the diode is reverse bias now.

Key details

- Obviously, so we get V output is equal to 0 over there.
- Then how to calculate the voltage across VD2?
- VD2 is 12 volts now.
- How it is 12 volts?
- Apply the KVL to the loop.
- We get VD2 is equal to E minus VD1 minus VD2, which is the 12 volts.
- Where Vout is equal to 0, yes, we'll be able to get, and VD1, we have 0.7 here.
- For ideal diode, it is 0, but 0.7, we get here 11.3 volts.
- Sorry, it is 11.3 volts we need to get.

- And Vout is equal to 0.
- I have here, which would be asked for one mark.
- See, it's not that the question could be asked for only four marks.
- A question could be asked for one mark also.
- Determine the current for each of the configuration.
- For this configuration, calculate the I.
- I is equal to, apply the.
- KVL, or first look at the negative voltage is connected here.
- When you apply negative, diode is reverse bias.
- When it's reverse bias, the current I is equal to 0.
- Since the diodes are connected in reverse here, the center branch will open here.
- The current will pass only through the resistor.
- Then coming to the few more numerical on parallel diode configuration.
- If the diodes are connected, one in same configuration, other in different configuration, one diode is ON, other diode is OFF.

Example or application from the lecture

- In the given question, they are asking us to calculate the VD2.
- C, calculate the current in the given configuration.
- Try to follow the slides and solve the numerical as per the given question, and we also will solve in the class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Parallel and Series- Parallel configurations transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Parallel and Series- Parallel configurations transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Parallel and Series- Parallel configurations transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Parallel and Series- Parallel configurations transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00026	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare...
SRC00027	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare...
SRC00028	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare...

C Revision and retrieval practice